

5-6-26

Test - 1

$$\frac{20}{20}$$

Sk Mishra.

5/6/26

✓. Good!

1) Decimal $\rightarrow$ A number system which uses 10 digits (0-9) and has the base 10 is called Decimal number system. ~~It is a 8-bit number system.~~

~~2) Binary~~ $\rightarrow$ A number system ~~which~~ uses 2 digits (0 and 1) and has the base 2 is called ~~Decimal~~ Binary number system. ~~It is a 2-bit number system.~~

~~3) Octal~~ $\rightarrow$ A number system which uses 8 digits (0-7) and has the base 8 is called Octal number system. ~~It is a 4-bit number system.~~

~~4) Hexadecimal~~ $\rightarrow$ A number system which uses 16 digits (0-9 and A-F) and has base 16 is called Hexadecimal number system. ~~It is a 8/16-bit number system.~~

2) The base (radix) of the hexadecimal number system is 16. ✓

$$\begin{array}{r} 2 \overline{) 25} \\ 2 \overline{) 12} \quad 1 \\ 2 \overline{) 6} \quad 0 \\ 2 \overline{) 3} \quad 0 \\ 1 \quad 1 \end{array}$$

$$(25)_{10} \rightarrow (11001)_{16}$$

$$4) (1010)_2$$

$$= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 8 + 0 + 2 + 0$$

$$= 10 \quad \checkmark$$

$$\therefore (1010)_2 \rightarrow (10)_{10}$$

$$5) \begin{array}{r|l} 2 & 15 \\ \hline 2 & 7 \quad 1 \\ 2 & 3 \quad 1 \\ 1 & 1 \end{array}$$

$$\therefore (15)_{10} \rightarrow (1111)_2$$

$$(1111)_2 \rightarrow \boxed{\text{F}}_{16} \quad (5)$$

$$\therefore \text{Ans} \rightarrow (F)_{16} \quad \checkmark$$

$$6) \begin{array}{r} 32168421 \\ (101101)_2 \end{array}$$

$$= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 32 + 0 + 8 + \boxed{4} + 0 + 1$$

$$= 45 \quad \checkmark$$

$$\therefore (101101)_2 \rightarrow \boxed{45}_{10} \quad \checkmark$$

$$7) (57)_8 = (101111)_2$$

$$\begin{array}{r|l} 2 & 57 \\ \hline 2 & 28 \quad 1 \\ 2 & 14 \quad 0 \\ 2 & 7 \quad 0 \\ 2 & 3 \quad 1 \\ & 1 \quad 1 \end{array}$$

$$\therefore 57$$

$$8) (3A)_{16} = (00111010)_2$$

3 2 1 0 1 0 1 0

$$(111010)_2 = 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 32 + 16 + 8 + 0 + 2 + 0$$

$$= (58)_{10}$$

$$\therefore (3A)_{16} = (58)_{10} \quad \checkmark$$

$$9) (156)_{10}$$

$$\begin{array}{r|l} 2 & 156 \\ \hline 2 & 78 \quad 0 \\ 2 & 39 \quad 0 \\ 2 & 19 \quad 1 \\ 2 & 9 \quad 1 \\ 2 & 4 \quad 1 \\ 2 & 2 \quad 0 \\ & 1 \quad 0 \end{array}$$

$$\begin{array}{r} 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \end{array}$$

$$\therefore (156)_{10} = (10011100)_2 \quad \checkmark$$

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$$16) \quad \overline{(11110110)}_2 = \text{F6}_{16}$$

$$11) a) \quad (1011)_2 + (1101)_2$$

$$\begin{array}{r} 1 1 \\ + 1 1 1 \\ \hline 1 1 0 0 \end{array}$$

$$\therefore \text{Ans} \rightarrow (11000)_2$$

$$b) \quad (11010)_2 - (10101)_2$$

$$\therefore \text{1st is complement of 2nd no.} = (01010)_2$$

$$\therefore \text{Adding 1st complement of 2nd no. to 1st no.} = \begin{array}{r} 1 0 0 \\ + 0 0 0 \\ \hline 1 0 1 0 \end{array}$$

Extra digit

Adding Extra digit to result

$$\begin{array}{r} 0 0 0 \\ + 1 \\ \hline 0 0 1 \end{array}$$

$$\therefore \text{Ans} \rightarrow (101)_2$$

$$c) \quad (101)_2 \times (11)_2$$

$$\begin{array}{r} 0 \\ \times 1 \\ \hline 0 \\ + 0 \\ \hline 1 \end{array}$$

$$\therefore \text{Ans} \rightarrow (1111)_2$$

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